

be like?

3. How will network security be addressed?

4. How will network elements be powered?

5. How will the system be commissioned and maintained?

6. What is the plan to manage interference?

7. What levels of interoperability and compliance are being sought?

System consideration. RF interference, blocked paths and node loss are the major factors that usually bring down a wireless link. So a WSN must protect itself against these to maintain very high end-to-end reliability. It must transparently adapt to changing environments, allowing long-term operation with zero-touch maintenance.

Infrequent transmissions, configuring the network on some other clearer channel in case of heavy network traffic, compression and digital filtering of sensor data prior to transmission to reduce data size and transmission time, and automatic retries are some of the means to avoid RF interference. Self-organising and self-healing intelligence, and routing discovery are a few techniques of self-adaptation to changing environments such as blocked paths and node failures.

Network topology. There are several types of network configurations that can be implemented with wireless technology: point-to-multipoint (star), tree and mesh.

Point-to-multipoint (star) networks are the simplest to implement and require the least amount of code for setup and control. These are typically limited in the number of nodes and coverage.

Tree networks allow more nodes; these can cover a larger area than point-to-multipoint networks. However, they suffer from latency effects that can cause critical node failure and shut the system down. Tree networks usually need more code to implement than multipoint systems.

Mesh networks represent the highest level of configuration and require the most network-level code. These can

self-heal critical node failures, but are the most difficult to design and implement. Mesh networks are more reliable than star and cluster tree/hybrid networks because these provide multiple paths for a message to traverse the network.

It is always advisable to choose the simplest network for the task at hand.

Network security. WSN designers must employ strong algorithms for data protection in the network. The adoption of the Advanced Encryption Standard (AES) with a 128-bit key length ensures data integrity and resistance to hacking. AES security may be implemented in software, although a dedicated hardware encryption processor provides a better solution as this minimises software overheads and allows faster encryption/decryption processing. This is important for sensor nodes that need to spend the shortest time possible awake, which is when they consume most power.

Power consideration. Most of the sensor nodes are battery-driven. Prolonged battery life is achieved by minimising the duration of time for which the radio is 'on,' which is when most power is consumed. A small packet size and infrequent transmission also help to achieve this.

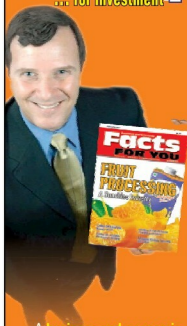
Currently, according to GreenPeak Technologies (Utrecht, the Netherlands), the aim is to develop chips and modules for sensor applications that would operate without power cabling or batteries. The requisite technologies are an ultra-low-power wireless transceiver and sensor interface design with efficient power-up and power-down modes that dramatically reduce power consumption; an energy-harvesting interface that lets the modules use power provided by external solar, electromagnetic and piezoelectric transducers; and a mesh technology that lets designers create extended sensor networks without the need for battery-powered or cabled routing nodes.

Commissioning and maintenance. Commissioning encompasses a wide range of tasks, including a survey of the radio and physical environment,

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